MIL-E-1/1036D
1 FEB 1976
SUPERSEDING
MIL-E-1/1036C
15 March 1962

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, MAGNETRON

TYPE 7529

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Pulsed, integral magnet, tunable 2700 to 2850 MHz, 3.5 MW peak power output, liquid cooled.

ABSOLUTE RATINGS:

Independent

Parameter:	If	epy	tk	VSWR	T(anode)	T(bushing)	Output pressurization
Unit:	A	kv	sec	----	°C	°C	psia
Maximum:	82 (Note 2)	70	---	1.5:1 (Note 4)	100 (Note 5)	150 (Note 5)	55 (Note 6)
Minimum:	68	--	180	----	---	---	45

Dependent

Parameter:	ib	Pi	pi	Du	tpc
Unit:	a	kW	Mw	--	μs
Maximum:	130	5.75	8.0	0.001	2.5
Minimum:	85	----	---	-----	1.0 (Note 8)

PHYSICIAL CHARACTERISTICS:

Dimensions:	See Figure 1	Weight:	66 pounds, approximate
Mounting position:	See Note 12	Cooling:	Liquid coolant (See Note 10)
Mounting support:	Brackets on magnet	Cathode:	Unipotential
Magnet isolation:	See Note 11		

TEST CONDITIONS:

Parameter:	If	If	Ib	VSWR	tk	tpc	tfc	tfv	trv	trc	Du
Unit:	A	A	mAdc	----	sec	μs	μs	μs	μs	μs	—
	Preheat	Test									
Maximum:	80	76	--	----	300	2.2	0.6	2.5	0.5	0.35	—
Nominal:	--	--	83	1.1:1	---	---	---	---	---	---	0.0008
Minimum:	75	70	--	----	---	1.8	---	---	0.3	---	--
							(See Note 1)				

GENERAL:

Qualification - Required

TEST FREQUENCY	
F	MHz
F_1	2700 ±5
F_2	2775 ±5
F_3	2850 ±5

7529

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METHOD	REQUIREMENT OR TEST	CONDITIONS	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Qualification inspection					
4027	Temperature coefficient	T(anode) = 40° to 70°C F_1, F_2 and F_3; See note 5	$\Delta F/\Delta T$	---	0.30	MHz/°C
4223	Mechanical tuning fatigue	T = 25° ±3°C; See note 13	---	500	---	cycles
4306	Pulse voltage	F_1 and F_3	epy	57	67	kv
4311	Frequency pushing figure	F_1, F_2 and F_3 See note 14	$\Delta F/\Delta ib$	--	0.050	MHz/A
1042	Shock	15G, 11 ms, three planes one direction	---	--	--	---
---	Cooling (anode)	T(coolant inlet) = approx 25°C; water coolant; See notes 5 and 15	---	--	--	---
		Flow = 0.8 gpm	ΔT	---	60	°C
			ΔP	---	0.18	psi
		Flow = 1.3 gpm	ΔT	---	51	°C
			ΔP	---	0.46	psi
		Flow = 1.8 gpm	ΔT	---	45	°C
			ΔP	---	0.85	psi
	Quality conformance Inspection, part 1					
4003	Pressurizing	55 psia min, output	---	---	---	---
---	Coolant flow	ΔP = 0.325 psi See note 15	---	1.0	---	gpm
---	Jacket pressure (anode and tuner)	See note 17	---	---	---	---
4223	Tuner torque	T(anode and tuner) = 25°C; See note 7	Torque	---	40	in.-oz
1261	Filament voltage	If = 76A; tk = 600 sec min	Ef	6.5	8.8	V
4223	Mechanical tuning range	T(anode) = 35° ±20°C	---	---	---	---
		High frequency	F	2850	---	MHz
		Low frequency	F	---	2700	MHz
4303	Heater-cathode warmup time	If = 78A; See notes 2 and 3	tk	---	300	sec

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METHOD	REQUIREMENT OR TEST	CONDITIONS	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1 continued					
4304	Pulse characteristics	tpc = 1.8 min to 2.2 μs max ; Du = 0.00072	---	---	---	---
4306	Pulse voltage	F_2	epy	57	67	kv
4250	Power output	t = 300 min	Po	2,520	---	W
4308	RF Bandwidth (1)	F_1, F_2 and F_3 See note 18	BW	---	1.5 tpc	MHz
4315	Stability	F_1, F_2 and F_3; See notes 12, 20 and 21	MP	---	1.0	%
	Quality conformance inspection, part 2					
1031	Low frequency vibration	No voltages	---	---	---	---
4308	RF Bandwidth (2)	Ib = 87 to 75 mAdc; See notes 18 and 22	BW	---	1.5	MHz
4310	Frequency pulling figure	F_1, F_2 and F_3	ΔF	---	20	MHz
	Quality conformance inspection, part 3					
----	Life test	Group D	---	---	---	---
4551	Intermittent life-test	See notes 23, 24 and 25	---	45	---	cycles
	Life-test end points:					
4250	Power output	F_1, F_2 and F_3	Po	2,000	---	W
4308	RF Bandwidth (1)	F_1, F_2 and F_3	BW	---	3.0	MHz
4315	Pulse stability	F_1, F_2 and F_3	MP	---	1.0	%

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NOTES:

1. This test shall be conducted for only one set of conditions within the limits specified for the test condition specified. The pulse characteristics shall be measured as follows:

   tfc and tfv - 0 to 85 percent point

   trc and trv - 20 to 85 percent point

   No spike ripple shall exceed ±7 percent of the average peak value of the voltage pulse. Inverse voltage shall not exceed 35 percent of the forward voltage. There shall be no negative post-pulse voltage.

2. The maximum value specified is for the nonoscillating condition. Heater surge current shall not be allowed to exceed 100 amperes.

3. The heater voltage may be reduced during operation at average anode current levels above 75 mAdc. The pertinent value of heater current must be maintained within ±3.5 percent of the recommended value.

4. Frequency skipping or unstable operation may be encountered at some phase positions when the mismatch occurs at the end of a "long" line.

5. The cathode, anode, and tuner temperatures shall be measured at points r, s, and t indicated on figure 1. Temperatures shall be measured after thermal equilibrium has been reached.

6. During operation, the gas used in pressurization shall provide insulating properties at least equal to that of clean, dry air at the pressures indicated.

7. The tuner drive mechanism shall not be set against either mechanical stop.

8. The characteristics of the applied pulse must be those which result in proper starting of oscillation. The rate of pulse voltage rise, the percentage of pulse voltage ripple, and the rate of the pulse voltage fall are among the more important considerations. Negative post pulse voltage oscillations may cause leakage current and noise.

9. The cathode terminal and high-voltage bushing shall be insulated by immersion in oil having the equivalent electrical characteristics of Federal Specification VV-I-530.

10. Coolant pressure shall not exceed 55 psia, but sufficient flow and pressure of coolant shall be maintained to provide cooling of the anode and tuner to specified temperature limits. The tuner shall be cooled by the equivalent of a water flow of 0.25 gallons per minutes (gpm) at 40°C maximum.

11. In handling and mounting the magnetron, care shall be exercised to prevent demagnetization. Ferromagnetic materials shall not be brought within 12 inches of the tube; energized magnets shall not be brought within 18 inches of the tube.

12. The tube shall be mounted with the cathode vertical within 15°.

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NOTES: -Continued

13. Starting from the mechanical stop, the tuner mechanism shall be rotated between 30 and 90 turns clockwise, counted at point of drive, to complete one excursion. A cycle shall consist of two complete excursions, each in opposite directions. Upon completion of this test, the tube shall be tested to and shall meet Power Output requirements.

14. The anode current shall be varied between 75 and 87 mAdc at a minimum rate of 50 cps.

15. The temperature rise of the anode above that of the coolant temperature at the inlet (ΔT) shall not exceed the limit specified for this test. The pressure drop across the anode coolant jacket (ΔP) shall not exceed the limit specified for this test.

16. The AQL for the combined defectives for attributes in Quality Conformance Inspection, part 1 excluding inoperatives and mechanical, shall be 1 percent.

17. With 55 psia of the liquid coolant applied at the input of jackets, and with the output of jackets plugged tight, there shall be no evidence of leakage after pressure has been applied for at least 1 minute.

18. The rf bandwidth shall be within the limits specified when a VSWR of 1.5:1 is introduced in the load at a distance of approximately 0.6 meter from the magnetron coupling flange, with the phase being adjusted for the widest spectrum.

19. Observations shall be made at the load phase position corresponding to maximum instability.

20. A missing pulse is defined as a pulse whose energy does not exceed 70 percent of the energy of a normal pulse at ± 1.0 percent of the test frequency. No more than the specified percentage of missing pulses shall be ascertained during this test, as counted during the last 5 minutes of a test period not exceeding 15 minutes.

21. This test shall be performed at the conclusion of the holding period.

22. The rf bandwidth shall be within the specified limit when the current is varied throughout the specified current range.

23. The standing wave introducer shall be moved during the test so that operation is obtained for load phase positions corresponding to maximum power output, minimum power output, maximum frequency, and minimum frequency. The automatically driven standing wave introducer shall be cycled continuously through a line length that is approximately 1/2 wave length long, at a maximum rate of 4 cycles per hour.

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NOTES: -Continued

24. Power input to the tube during life test shall be cycled according to the following schedule:

    Preheat (heater) - 5 minutes

    Oscillate (RF) - 11.5 hours

    All voltages off - 25 minutes

25. During life test change frequency in 75 MHz increments at 24 hour time intervals

Custodians:
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Navy - EC
Air Force - 85

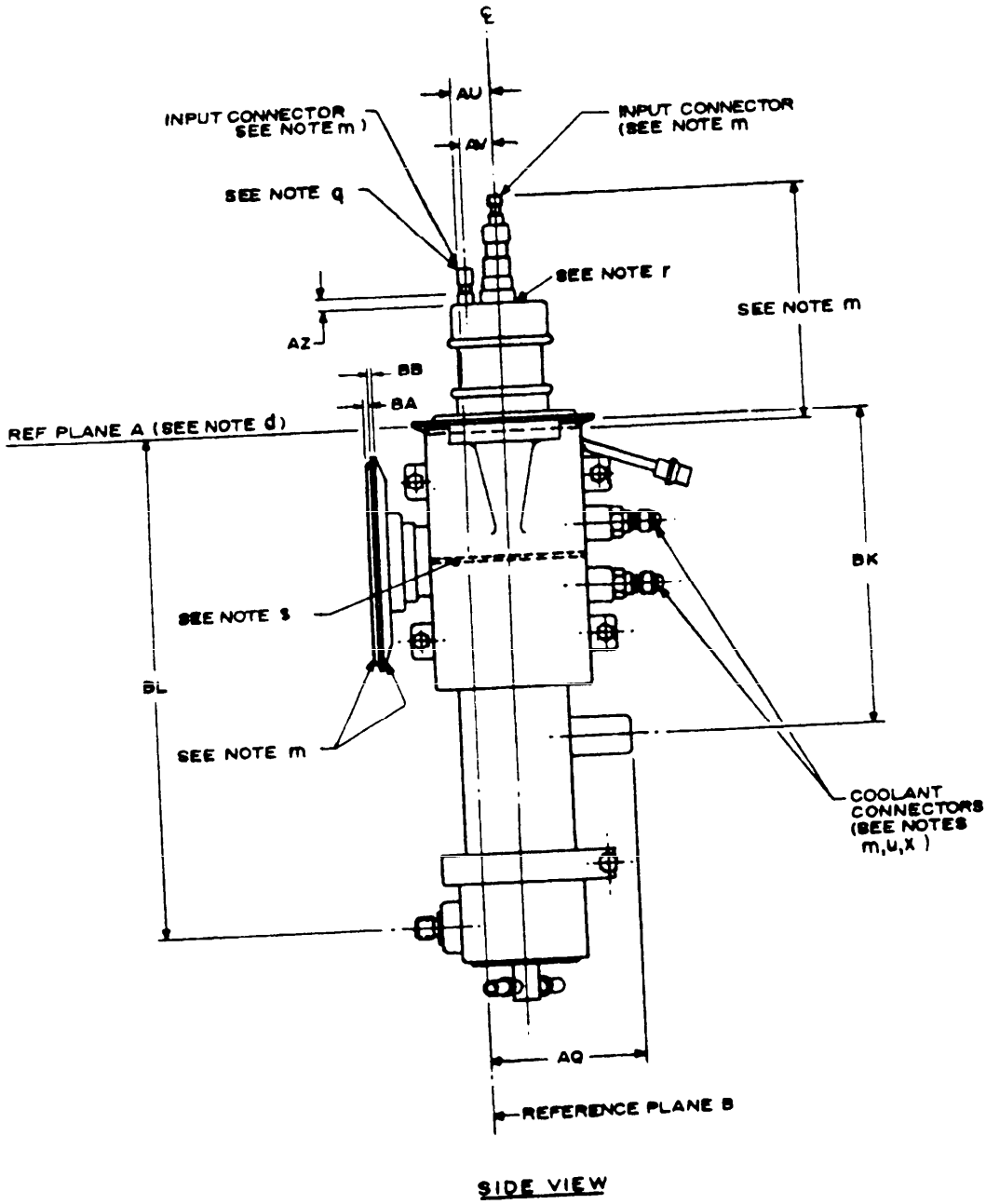
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Air Force - 80
DSA - ES

User activities:
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Navy - AS, OS, MC, CG, SH
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Preparing activity:
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SIDE VIEW

FIGURE 1 Outline drawing of electron tube type 7529

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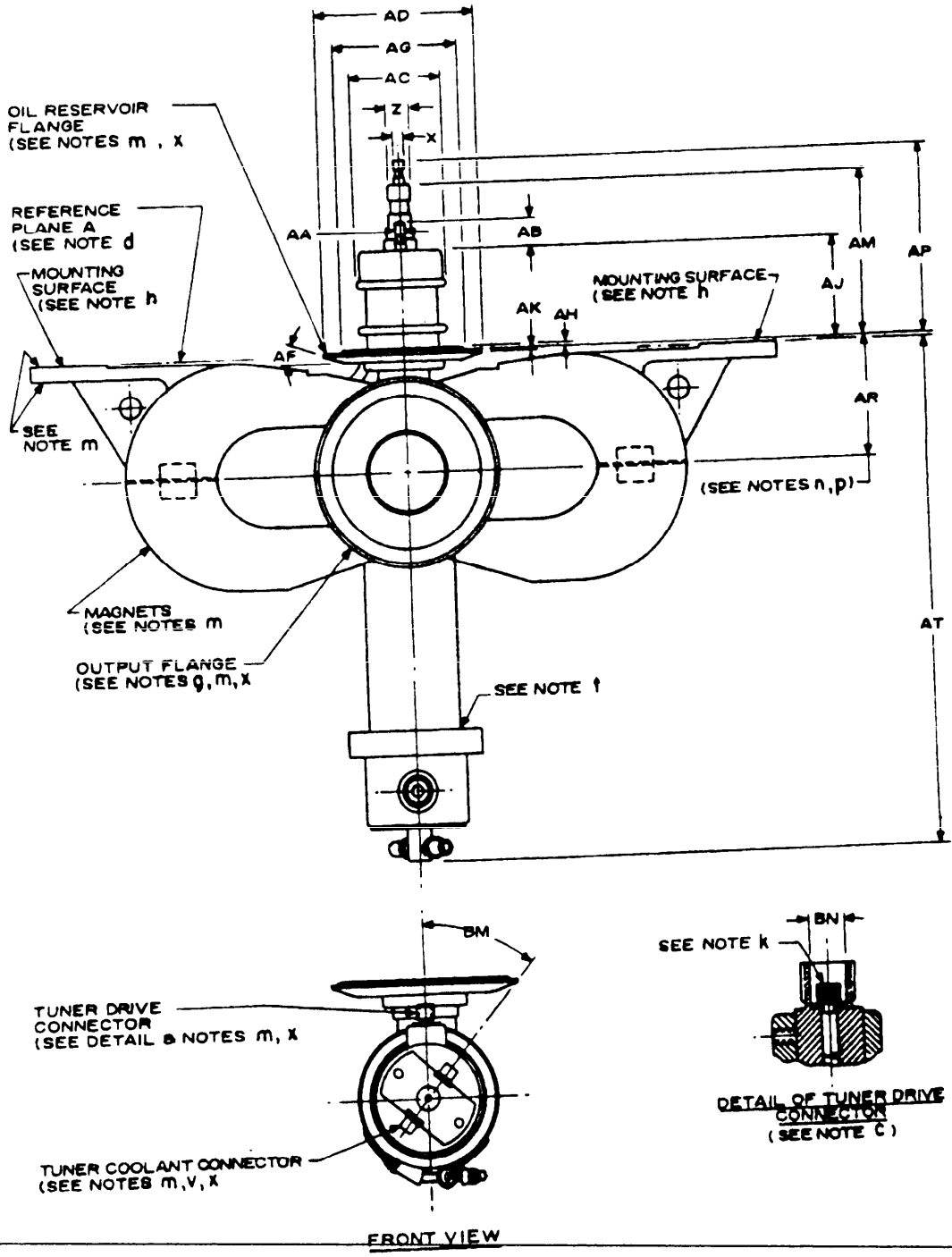
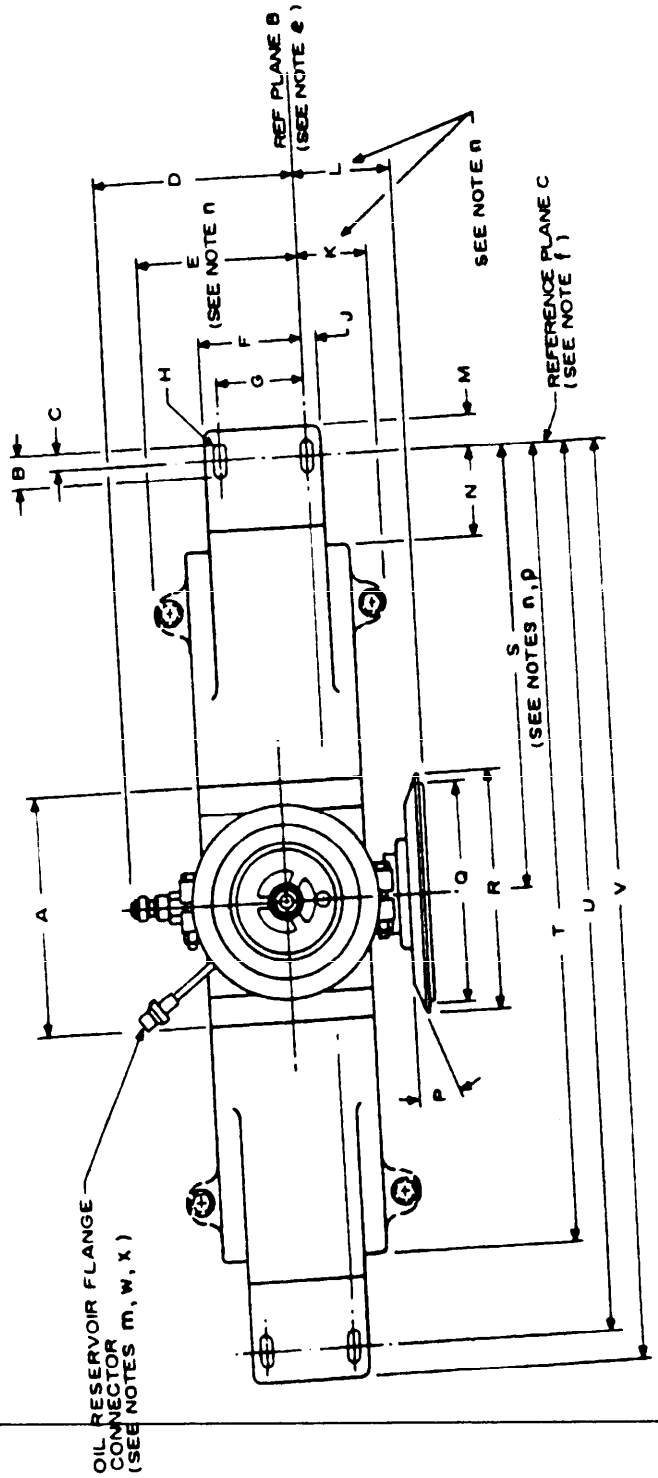


FIGURE 1 Outline drawing - continued

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TOP VIEW

FIGURE 1 Outline drawing - continued

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DIM	INCHES		MILLIMETERS	
	MIN	MAX	MIN	MAX
QUALITY CONFORMANCE INSPECTION, PART 1				
B	.860	890	21.84	22.61
C	425	450	10.80	11.43
H	190 R	210 R	4.83 R	5.33 R
L	1.961	2.211	49.81	56.16
S	9.662	9.962	245.41	253.03
U	19.437	19.815	493.70	503.30
V	-	20.625	-	523.88
AC	-	2.875	-	73.03
AJ	-	3.500	-	88.90
AM	-	5.000	-	127.00
AP	-	6.000	-	152.40
AR	3.250	3.500	82.55	88.90
AT	-	14.750	-	374.65
QUALITY CONFORMANCE INSPECTION, PART 2				
A	4.812	-	122.22	-
E	-	3.875	-	98.43
F	-	2.437	-	61.90
G	1.985	2.015	50.42	51.18
J	-	.437	-	11.10
K	-	1.812	-	46.02
M	-	.750	-	19.05
N	1.250	-	31.75	-
Q	5.025	5.051	127.64	128.30
R	5.285	5.315	134.24	135.00
T	-	18.500	-	469.90
X	-	.344	-	8.74
AA	-	.312	-	7.92
AD	4.485	4.515	113.92	114.68
BN	.466	.472	11.84	11.99
REFERENCE DIMENSIONS (NOTE 6)				
D	4.917		124.89	
P	20°			
Z	.740		18.80	
AB	.875		22.23	
AG	3.485		88.52	
AH	.125		3.18	
AK	.063		1.60	
AQ	4.020		102.11	
AU	1.000		25.40	
AV	.850		21.59	
AZ	.250		6.35	
BA	.094		2.39	
BB	.063		1.60	
BK	9.765		248.03	
BL	12.200		309.88	
BM	40°			
AF	20°			

FIGURE 1 Outline drawing - continued

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NOTES:

a. All dimensions in inches, unless otherwise specified.
b. These dimensions are for information only and are not required for inspection purposes.
c. Total number of revolutions of small bevel gear, between limiting stops of tuning mechanism, shall be not fewer than 100. Ratio of bevel gears is 5 to 1.
d. Reference plane A is defined as that plane which is coincident with mounting surfaces.
e. Reference plane B is defined as that plane which is perpendicular to reference plane A and which passes through centers of two mounting holes as shown.
f. Reference plane C is defined as that plane which is mutually perpendicular to reference planes A and B and which passes through center of one mounting hole as shown.
g. Soldered joints of output section shall provide hermetic connection with output mating flange. Output flange is gasketed for pressurization.
h. These surfaces shall be coplanar within 0.032.
j. Tuner drive connector thread shall be 5/8-24 NEF-2A.
   Major diameter shall be not less than 0.6145.
   Pitch diameter shall accept 5/8-24 NEF-2 "GO" gage only.
k. Spline specifications: 14-1/2° pressure angle. 48 pitch. 12 splines. 0.250 pitch diameter.
m. All metal surfaces of tube except front and rear surfaces of oil reservoir flange, parts located above oil reservoir flange, front and rear surfaces of output flange, mounting surfaces of bracket, input connectors, coolant connectors, oil reservoir flange connector, and tuner drive connector shall be painted with heat resistant noncorrosive paint.
n. Includes angular and lateral deviations.
p. Refers to centerline of waveguide.
q. Common cathode connection.
r. Cathode stem temperature shall be measured at this point.
s. Anode temperature shall be measured at this point.
t. Tuner temperature shall be measured at this point, or as near this point as is practicable.
u. Connectors shall be Hansen type B3-T20, or equivalent. Two connectors required.
v. Connectors shall mate with Wiggins connector type P-1706, or equivalent. Two connectors required.
w. Connector shall mate with Wiggins connector type P-1707, or equivalent. One connector required.
x. Protective shipping covers shall be provided with tube for output flange, oil reservoir flange and parts located above oil reservoir flange, oil reservoir connector, tuner coolant connectors, and tuner drive connector.

Figure 1. Outline drawing. - Continued

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